
SEPHORA PRODUCT & CUSTOMER REVIEW ANALYT

The domain of the Project: Data Analytics & Artificial Intelligence

Course : Data Analytics

Team Mentors (and their designation):

Keerthana V. Mam (Senior Executive - Business Analyst Exl Service)

Sama Varshitha– G2 Data Analytics

Period of the project

July 2026 to September 2026

Declaration

The project titled **“Sephora products and customer reviews Analysis”** has been mentored by **Keerthana V. Mam**, organised by SURE Trust, from July 2026 to September 2026, for the benefit of the educated unemployed rural youth for gaining hands-on experience in working on industry relevant projects that would take them closer to the prospective employer. I declare that to the best of my knowledge the members of the team mentioned below, have worked on it successfully and enhanced their practical knowledge in the domain.

Team Members:

Ms.Sama Varshitha

Mentor's Name

Keerthana V.

Senior Executive

Business Analyst Exl Service

Prof. Radhakumari

Executive Director & Founder

SURE Trust

TABLE OF CONTENTS

- 1. Executive Summary 4
- 2. Introduction..... 5
- 3. Project Objectives..... 7
- 4. Methodology and Results..... 9
- 5. Social/Industry Relevance of the Project 15
- 6. Learning and Reflection..... 17
- 7. Future Scope 19
- 8. Conclusion.....21



1. Executive Summary

Data Analytics has become an important part of the modern retail and e-commerce industry. Large organizations collect huge amounts of data related to products, customers, reviews, prices, ratings, brands and customer preferences. Analysing this information helps organizations understand customer behaviour, improve products and make better business decisions..

The project “Sephora Product & Customer Review Analytics” focuses on analysing Sephora product and customer review datasets using Data Analytics techniques. The project combines two major datasets: a cleaned product catalogue dataset and a customer reviews dataset.

The product dataset contains information about products such as product name, brand, category, price, rating and customer engagement indicators. The reviews dataset contains customer-generated information such as product reviews, ratings, recommendation status and other available review-related attributes.

The main purpose of this project is to transform raw datasets into meaningful information through data cleaning, exploratory data analysis, statistical analysis and visualization.

The project follows a systematic data analytics workflow:

**Data Collection → Data Cleaning → Data Pre-processing → Exploratory Data Analysis
→ Data Visualization → Interpretation → Insights → Report Generation**

The analysis helps identify:

- Distribution of products across different categories.
- Brands having higher customer engagement.
- Distribution of customer ratings.
- Products receiving a large number of reviews.
- Average prices across different product categories.
- Customer recommendation patterns.
- General patterns in customer feedback.

The project provides practical experience in using Python-based data analytics techniques and demonstrates how raw business data can be converted into useful insights.

The analysis can further be extended into advanced applications such as customer sentiment analysis, product recommendation systems, predictive modelling and interactive dashboards.



2. Introduction

2.1 Background

The beauty and cosmetics industry is highly competitive and customer-driven. Companies offer thousands of products across categories such as skincare, makeup, fragrance, haircare and other beauty products.

Customers generate valuable data whenever they search for products, purchase products, rate products or write reviews. Product reviews are particularly useful because they provide direct feedback from customers about their experiences.

However, large datasets cannot be effectively understood simply by looking at individual records. Data Analytics provides methods for organizing, processing and analysing such datasets.

This project uses Sephora-related product and customer review datasets to demonstrate a practical Data Analytics workflow.

2.2 Problem Statement

The raw product and review datasets contain a large number of records and multiple attributes. Without proper processing, it is difficult to identify useful patterns.

The problem addressed by this project is:

How can Sephora product and customer review data be analysed to identify meaningful patterns in product categories, pricing, ratings, customer recommendations and brand engagement?

The project attempts to answer this question using descriptive analytics and visualization techniques.

2.3 Need for the Project

The project is important because data-driven decision-making is widely used in the retail industry. Analysis of product and review data can help businesses:

1. Understand customer preferences.
2. Identify popular product categories.
3. Monitor customer satisfaction.
4. Compare product performance.
5. Understand brand engagement.
6. Identify highly reviewed products.
7. Analyse pricing patterns.
8. Improve customer experience.



2.4 Scope of the Project

The scope of this project includes:

- 1.Product data analysis.
- 2.Customer review analysis.
- 3.Data cleaning.
- 4.Exploratory Data Analysis.
- 5.Statistical summaries.
- 6.Category analysis.
- 7.Brand analysis.
- 8.Rating analysis.
- 9.Price analysis.
- 10.Customer recommendation analysis.
- 11.Data visualization.
- 12.Business interpretation.

The project is primarily focused on descriptive and exploratory analytics.

2.5 Limitations

The project has some limitations:

The analysis is based only on the supplied datasets.

The dataset may not represent the complete current Sephora catalogue.

Review data represents customer opinions and can contain subjective responses.

Descriptive analysis cannot establish causal relationships.

The project does not directly measure actual sales unless sales data is available.

The results should therefore be interpreted within the context of the supplied datasets..



3. Project Objectives

The main objective of the project is to analyse Sephora product and customer review data and extract meaningful insights.

Specific Objectives:

Objective 1: Data Collection

To collect and work with structured product and customer review datasets.

Objective 2: Data Cleaning

To identify and handle missing values, inconsistent formats, duplicate records and unsuitable data types where required.

Objective 3: Product Analysis

To analyse products according to:

- Brand
- Category
- Price
- Rating
- Customer engagement

Objective 4: Customer Review Analysis

To analyse customer reviews according to:

- Rating
- Recommendation status
- Review volume
- Product
- Available customer attributes

Objective 5: Category Analysis

To identify which product categories contain the highest number of products and compare their characteristics.

Objective 6: Brand Analysis

To identify brands with high customer engagement based on the available engagement indicators.

Objective 7: Price Analysis

To calculate and compare average product prices across different categories.

Objective 8: Visualization

To create charts and graphs that communicate analytical findings clearly.

Objective 9: Business Insights

To convert analytical results into useful observations that could support retail decision-making.



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

Objective 10: Future Development

To establish a foundation for future machine-learning, sentiment-analysis and recommendation-system applications.



4. Methodology and Results

The methodology used in the project consists of the following stages:

Step 1: Data Collection

Two datasets were used:

Dataset 1: products_cleaned.csv

This dataset contains product-level information.

Dataset 2: reviews.csv

This dataset contains customer review information.

Step 2: Data Understanding

The datasets were inspected to understand:

1. Number of records.
2. Number of columns.
3. Column names.
4. Data types.
5. Missing values.
6. Duplicate records.
7. Numerical and categorical variables.

Step 3: Data Cleaning

Data cleaning is an important step because inaccurate or inconsistent data can produce incorrect results.

The following activities were considered:

1. Checking missing values.
2. Checking duplicate records.
3. Converting columns into appropriate data types.
4. Handling date fields.
5. Checking rating values.
6. Checking numerical values.
7. Standardizing categorical fields where necessary.

Step 4: Exploratory Data Analysis

Exploratory Data Analysis was performed to understand the characteristics of the datasets.

The analysis included:

1. Frequency analysis.
2. Group-by analysis.
3. Average calculations.
4. Rating distribution.
5. Category distribution.
6. Brand analysis.
7. Price comparison.
8. Review-volume analysis.



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

Step 5: Data Visualization

Charts were created to communicate the results.

Examples include:

1. Product category distribution.
2. Top brands by engagement.
3. Review rating distribution.
4. Average price by category.

Step 6: Interpretation

The visualizations and statistical results were interpreted to identify useful patterns.

Step 7: Reporting

The final findings were organized into a structured project report.

4.2 Technologies Used

Python

Python was used as the primary programming language for data analysis.

Python is suitable for this project because it provides powerful libraries for:

1. Data manipulation.
2. Statistical analysis.
3. Visualization.
4. Data cleaning.

Pandas

Pandas was used for:

1. Reading CSV files.
2. Data manipulation.
3. Filtering records.
4. Grouping data.
5. Calculating averages.
6. Generating summary statistics.

NumPy

NumPy can be used for numerical operations and mathematical calculations.

Matplotlib

Matplotlib was used to create visualizations such as bar charts and rating distributions.

Microsoft Word

Microsoft Word was used to prepare the final project documentation.



4.3 Dataset Description

Product Dataset

The product dataset contains information about Sephora products.

Important fields include:

1. Product ID
2. Product Name
3. Brand Name
4. Primary Category
5. Price
6. Rating
7. Loves Count
8. These fields allow the project to study product availability, pricing, ratings and customer engagement.

Review Dataset

The review dataset contains customer feedback information.

Important fields include:

1. Product ID
2. Product Name
3. Rating
4. Recommendation indicator
5. Review text
6. Submission time
7. Customer-related fields where available

The review dataset allows customer feedback and rating behaviour to be analysed.

4.4 Data Analysis

The supplied product dataset contains **product records**, while the review dataset contains **customer review records**. The analysis was performed separately and then interpreted together.

Product Category Analysis

Products were grouped according to their primary category.

This helps determine which categories have greater representation in the catalogue.

The major categories include areas such as:

1. Skincare
2. Makeup
3. Hair
4. Fragrance
5. Gifts

Other available categories

The category analysis provides an overview of the product assortment.

Brand Analysis

Products were grouped by brand and engagement indicators were aggregated.

One useful indicator is **loves_count**, which represents customer engagement recorded in the product dataset.

Brands with higher aggregate engagement can be identified for further analysis.

However, loves_count should not be interpreted as actual sales because the dataset does not establish that relationship.



4.5 Rating Analysis

Rating analysis is one of the most important parts of the project.

Ratings help understand customer perception of products.

The project examines:

1. Average product rating.
2. Average review rating.
3. Number of 1-star reviews.
4. Number of 2-star reviews.
5. Number of 3-star reviews.
6. Number of 4-star reviews.
7. Number of 5-star reviews.

A rating distribution chart makes it easier to understand the overall customer rating pattern.

If a large proportion of reviews have high ratings, this indicates that the supplied review dataset contains predominantly positive customer feedback.

However, this should not automatically be interpreted as the satisfaction level of all Sephora customers because the dataset represents only the available records.

4.6 Recommendation Analysis

The review dataset contains information indicating whether customers recommended a product.

The recommendation rate can be calculated using:

Recommendation Rate = Recommended Reviews / Total Reviews × 100

This metric provides a simple indicator of customer willingness to recommend products.

Recommendation analysis can be useful for identifying products or categories that have stronger customer approval.

4.7 Price Analysis

Price is an important factor in retail analytics.

The project compares average product prices across primary categories.

For each category:

1. **Average Price = Total Price of Products / Number of Products**
2. Price analysis can help identify:
3. Premium categories.
4. Lower-priced categories.
5. Categories with greater price variation.
6. Potential differences between categories.

Price should be analysed together with product mix because an average price can be influenced by premium brands, larger product sizes, gift sets and other catalogue characteristics.



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

4.8 Review Volume Analysis

Review volume provides an indication of how much customer feedback is available for a product.

Products can be ranked according to the number of reviews.

Highly reviewed products can be selected for deeper analysis.

For example, a product with many reviews may be useful for:

Sentiment analysis.

Customer feedback analysis.

Rating analysis.

Recommendation analysis.

Review volume alone does not necessarily mean that a product has higher sales.

4.9 Project Architecture

The project architecture can be represented as:

Raw Product Dataset

↓

Raw Review Dataset

↓

Data Loading

↓

Data Cleaning & Validation

↓

Data Pre-processing

↓

Exploratory Data Analysis

↓

Statistical Analysis

↓

Data Visualization

↓

Insight Generation

↓

Final Report

This architecture provides a simple and systematic approach for converting raw data into useful information.

RESULTS AND FINDINGS

The analysis generated several important observations.

Finding 1: Category Distribution

The product catalogue contains a large number of products across multiple beauty categories.

Skincare and Makeup are among the major categories represented in the dataset.

This demonstrates the broad product assortment present in the supplied catalogue.



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

Finding 2: Customer Ratings

Customer ratings are concentrated towards the higher rating values.

This means that positive ratings represent a substantial portion of the review dataset.

Finding 3: Customer Recommendations

The recommendation field provides another indicator of customer response.

A high recommendation rate suggests that customers frequently indicate that they would recommend products.

Finding 4: Brand Engagement

The loves_count field can be aggregated by brand to identify brands with higher recorded customer engagement.

This can help prioritize brands for additional analysis.

Finding 5: Price Differences

Average product prices vary across categories.

Premium categories may have higher average prices because of differences in brands, product types, sizes and product positioning.

Finding 6: Review Volume

Some products have substantially more reviews than others.

Products with high review volume provide richer data for future customer sentiment analysis.



5. Social/Industry Relevance of the Project

5.1 Industry Relevance

Data Analytics is widely used in the retail and e-commerce industry.

Retail companies can use analytics to understand:

1. Customer preferences.
2. Product popularity.
3. Customer satisfaction.
4. Pricing.
5. Brand performance.
6. Product assortment.
7. Customer feedback.

The Sephora project demonstrates these concepts using real-world-style product and review data.

5.2 Business Applications

The analysis can support:

Product Management

Businesses can understand which categories contain large numbers of products and which products receive significant customer attention.

Customer Experience

Review and recommendation analysis can help organizations understand customer feedback.

Marketing

Brand engagement information can be useful for understanding customer interest.

Pricing

Category-level price analysis can support pricing research.

Product Recommendations

Customer ratings and product attributes can be used as inputs for recommendation systems.



5.3 Social Relevance

The project is also relevant from an educational perspective.

Students gain practical experience in:

1. Data handling.
2. Programming.
3. Analytical thinking.
4. Visualization.
5. Problem-solving.
6. Business interpretation.

Such skills can improve employability in areas such as:

1. Data Analyst
2. Business Analyst
3. Junior Data Scientist
4. BI Analyst
5. Reporting Analyst
6. Python Data Analyst

.



6. Learning and Reflection

6.1 Technical Learning

During the project, the following technical skills were developed:

Python

Improved understanding of Python for practical data analysis.

Pandas

Learned how to:

1. Read CSV files.
2. Inspect datasets.
3. Filter data.
4. Group records.
5. Calculate statistics.
6. Work with missing values.

Data Cleaning

Learned that data preparation is an essential part of any analytics project.

Data Visualization

Learned how charts can make complex datasets easier to understand.

Exploratory Data Analysis

Learned how to investigate datasets before building advanced models.

6.2 Analytical Learning

The project improved understanding of:

- Descriptive statistics.
- Average and frequency calculations.
- Category comparison.
- Rating distribution.
- Customer recommendation metrics.
- Data interpretation.

An important learning was that numbers alone are not sufficient. The analyst must understand what each metric represents before making conclusions.

6.3 Teamwork Learning

The project also provided experience in:

- Dividing project responsibilities.
- Discussing analytical findings.
- Reviewing results.
- Preparing documentation.
- Presenting information clearly.
- Meeting project requirements.



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

6.4 Overall Reflection

This project provided practical exposure to a complete data analytics workflow.

Initially, raw datasets can appear difficult to understand because they contain many records and columns.

By systematically cleaning and analysing the data, meaningful patterns can be identified.

The project helped develop confidence in working with datasets and improved understanding of how Data Analytics is applied to real-world business problems.



7. Future Scope

The project can be extended in several ways.

7.1 Sentiment Analysis

The review text can be analysed using Natural Language Processing techniques.

Reviews can be classified as:

1. Positive
2. Negative
3. Neutral

This would provide deeper insights than numerical ratings alone.

7.2 Product Recommendation System

A recommendation system can be developed using:

1. Product categories.
2. Product ratings.
3. Customer preferences.
4. Review information.
5. Product attributes.

The system could recommend products similar to products liked by customers.

7.3 Machine Learning

Machine-learning models can be developed to predict:

1. Recommendation probability.
2. Customer rating.
3. Product popularity.
4. Review sentiment.



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

7.4 Interactive Dashboard

An interactive dashboard can be created using:

1. Power BI
2. Tableau
3. Streamlit

The dashboard could contain:

1. Category filters.
2. Brand filters.
3. Price analysis.
4. Rating charts.
5. Review trends.
6. Recommendation statistics.

7.5 Time-Series Analysis

The review submission date can be used to study changes in review volume over time.

This can help identify:

1. Increasing customer interest.
2. Seasonal patterns.
3. Changes in review activity.

7.6 Advanced Customer Analytics

If additional customer and transaction data becomes available, future analysis could include:

1. Customer segmentation.
2. Purchase behaviour.
3. Customer lifetime value.
4. Churn prediction.
5. Personalized recommendations.



8.Conclusion

The **Sephora Product & Customer Review Analytics** project demonstrates how Data Analytics techniques can be applied to product and customer review datasets.

The project followed a structured workflow beginning with data loading and cleaning, followed by exploratory analysis, statistical calculations and visualization.

The analysis examined important areas including:

1. Product categories.
2. Brands.
3. Product prices.
4. Product ratings.
5. Customer review ratings.
6. Customer recommendations.
7. Review volume.
8. Brand engagement.

The project demonstrates that data analytics can transform large and complex datasets into meaningful information.

The experience gained through this project is useful for developing practical skills required for entry-level Data Analytics roles. It also provides a foundation for advanced applications involving Artificial Intelligence, Machine Learning, Natural Language Processing and recommendation systems.

In conclusion, the project successfully demonstrates the importance of data-driven analysis in the retail industry and provides a strong foundation for future development